DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

Public reporting burden for this collection of information is estimated to average 110 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503.

1. TITLE	2. IDENTIFICATION NUMBER
Flight Termination System Report (FTSR)	DI-SAFT-80183A

3. DESCRIPTION / PURPOSE

3.1 The FTSR provides a detailed description of the Flight Termination System (FTS). It is the medium through which FTS approval is obtained from the launch range.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
950809	F/AFDTC-SE		

7. APPLICATION / INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

7.2 This DID is applicable to all missile and space contracts that include direct support or management of missile launches and other munitions which require an FTS.

7.3 This DID supersedes DI-SAFT-80183.

APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		F7160

10. PREPARATION INSTRUCTIONS

10.1 Format. Contractor format is acceptable.

10.2 Content. The FTSR contains technical information on the FTS. When applicable, previously furnished documentation shall be referenced rather than incorporated into the report. The FTSR includes the following information:

10.2.1 Introduction. Scope and purpose.

10.2.2 General system description. This section contains a brief and general description of the FTS and includes a block diagram showing the location of all FTS components on the vehicle, including interfaces with other systems.

10.2.3 Detailed system description. This section contains the complete and detailed narrative description of all the major components of the FTS. The following items are included in this section:

a. Narrative description.

(1) The complete and detailed description of the FTS operation including all possible scenarios and discussion of how FTS components function at the system and piece-part levels.

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DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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Block 10, Preparation Instructions (Continued)

(2) A complete and detailed description of each FTS component and how it functions, including references to specifications and schematics, mechanical and piece-part specifications, and operating parameters.

b. Detailed schematics of the complete FTS showing component values (resistance, capacitance, wattage), tolerances, shields, grounds, connectors and pin numbers, and telemetry pickoff points. The schematics include all vehicle components and elements which interface with or share common use with the FTS. All pin assignments shall be accounted for. Results of the bent-pin analysis.

c. Drawings showing the location of all FTS system components on the vehicle. Description of element siting, mounting (attach points), and cable routing (for physical isolation). Includes a description of electrical connectors and electrical isolation of the FTS. Include an illustrated parts breakdown of all mechanically operated FTS components.

d. Antenna link analysis and battery load analysis.

10.2.4 Environmental testing. This section references all qualification, acceptance, and certification test plans, procedures, and reports. This section includes the following data:

a. The maximum predicted flight loads (per axis for three axis) for all anticipated environmental forces (for example, shock, vibration, acceleration and thermal) for each FTS component. Include a summary of the analyses or measurements used to derive the maximum predicted environments for each component.

b. A matrix of the actual qualification and acceptance test levels used for each component in each test versus the predicted flight levels for each environment.

c. A clear identification of those components qualified by similarity, analysis or a combination of analysis and test.

10.2.5 Failure analysis and corrective action. This section contains all failure analyses, reports and corrective actions for all FTS major components which fail to meet specifications, tolerances or test procedure requirements.

10.2.6 Modifications. This section includes all proposed and final modifications to an approved FTS, associated equipment, component identification, test procedures, or any changes affecting the configuration or integrity of the FTS.

10.2.7 Reliability analysis. This section includes all analysis and data to support the reliability requirements for the FTS.

a. A discussion of how the FTS meets the design requirements of single-fault tolerance and reliability design goal.

b. The results of fratricide and sneak circuit analyses when required

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Block 10, Preparation Instructions (Continued)

c. Include the Failure Modes, Effects, and Criticality Analysis (FMECA) when required.

10.2.8 Installation and checkout. This section includes all procedures for installation and checkout of all FTS components and a description of all equipment used during the checkout, to include a wiring diagram depicting any interfaces to the FTS.

10.2.9 Waivers. This section contains all required and existing waivers, technical agreements, and understandings concerning FTS.

10.2.10 Telemetry design data. Includes the following data:

a. A complete and detailed description of the FTS telemetry system and how it functions, including general specifications and schematics.

b. A listing of typical telemetry calibration data for each safety channel.

c. A tabulation of measurement accuracy, sampling rate, and time delay for each telemetry measurement used for safety purposes.

d. A set of single-thread wiring diagrams showing the end-to-end schematic for each safety monitoring channel from the pickoff point to telemetry input, ground support equipment, or safety console readout. Identify all connectors and pin numbers, components, and wire sizes.

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